Pretraining-Based Relevance-Aware Visit Similarity Network for Drug Recommendation

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***Abstract*—Drug recommendation based on electronic health record (EHR) is fundamental to effective disease treatment. Similar to commercial sequence-based recommendation systems, the accuracy of drug recommendation largely depends on precise patient modeling. However, patient modeling is more complex, as it not only requires sequence modeling of patient's disease course, but also needs to refer to the information of patients with similar medical medication. In EHR data, many patients have only one visit record, and the similarity between patients is often vague and unclear, which may cause noise and ambiguity. This leads to significant challenges for the drug recommendation field, especially when patient records are sparse or when patient similarity is vague. To address the above challenges, we propose RaVSNet (Relevance aware Visit Similarity Network), which improves drug recommendation by leveraging both longitudinal and transversal visit similarity and integrating medical relevance knowledge. RaVSNet utilizes multi-dimensional visit information similar to the patient's current visit as a reference, and employs a relevance-aware network to explicitly model the matching relationships between medical conditions and medications. Additionally, RaVSNet designs a general pretraining framework specifically for drug recommendation, including two tasks, Medication Sequence Reconstruction (MSR) and Causal Effect Inference (CEI), to discover the deep connections between medical information and medications. Experimental results on two public EHR datasets, MIMIC-III and MIMIC-IV demonstrate that the proposed algorithm outperforms state-of-the-art methods, yielding more accurate drug recommendation combinations, and the proposed general pretraining framework can be seamlessly integrated into most drug recommendation methods to achieve performance improvements.**

***Index Terms*—Data mining, electronic health records, intelligent healthcare management, medication recommendations, patient modeling.**

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The implementation is available at: https://github.com/SCUT-CCNL/RaVSNet.

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I. Introduction

THE integration of Electronic Health Records (EHRs) with deep learning models has significantly advanced the landscape of intelligent healthcare [1], providing unprecedented opportunities for improving diagnosis, prognosis, and treatment planning [2]. The widespread adoption of EHR systems has enabled the structured and large-scale collection of longitudinal patient data, including diagnoses, procedures, medications, and laboratory tests [3]. With the rapid progress of artificial intelligence (AI), particularly in deep neural networks [4], meaningful patterns can be extracted from these high-dimensional and heterogeneous data sources. Consequently, AI-driven techniques have catalyzed numerous transformative healthcare applications [5], including clinical decision support [6], medical image analysis [7], and disease prediction [8], shaping the future of precision medicine.

Despite notable success in domains such as medical imaging [9], the application of AI in medication recommendation remains relatively underdeveloped [10], [11]. Unlike conventional recommendation tasks [12], [13], medication recommendation is inherently more complex, as it requires generating a combination of medications for a given patient condition while considering safety constraints, such as drug-drug interactions (DDIs) [14]. Early drug recommendation systems predominantly leveraged traditional sequence-based recommendation techniques, aiming to predict subsequent actions or items based on the patient's current state [13]. Representative methods such as LEAP [15] offer a tailored treatment plan by focusing on immediate diagnoses, yet they fail to capture a patient's health trajectory over time, which is crucial in chronic and complex conditions. To address this, longitudinal models have emerged as a promising solution, such as DEPOT [16], MoleRec [17] and StratMed [18], which aggregate the patient's current health status and medical history information, using Recurrent Neural Networks (RNNs) to learn patient representations. This allows the model to capture the patient's current health state, which is then fed into the predictor to recommend the most suitable drug combinations.

Providing accurate and safe medication recommendations for patients remains a complex challenge, driven by several critical issues. First, exploring visit similarities within Electronic Health Records (EHR), especially across both longitudinal and transversal dimensions, presents a substantial hurdle. Most existing models often aggregate all historical visits without

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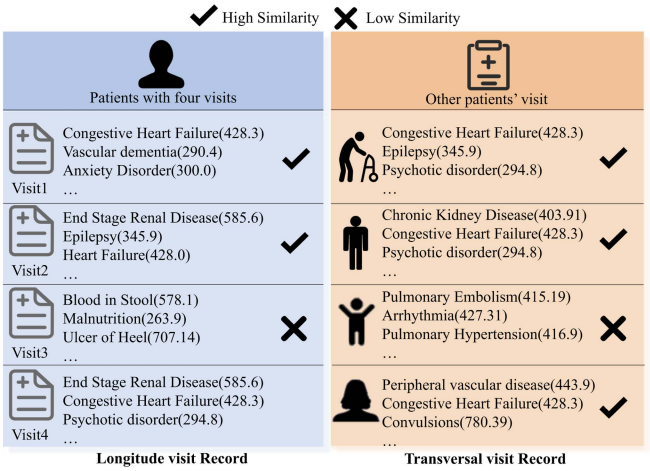


Fig. 1. Example: A patient has four visit records, and the most recent visit may not be similar to previous ones. However, visit records from other patients may be more similar to the patient's most recent visit.

considering their clinical relevance to the current condition, which may introduce noise or redundancy in acute cases where past medical events are unrelated [19], [20]. In cases where a patient presents with an acute illness, their historical records may not be relevant, posing challenges for existing longitudinal-based drug recommendation models. For example, Fig. 1 illustrates a simple case where the patient's fourth visit clearly differs from the second visit in terms of their medical condition. Blindly aggregating all the patient's historical visit records could introduce noise, thereby affecting the accuracy of the model's recommendations. Furthermore, for patients visiting for the first time or presenting with acute conditions, the lack of direct historical data compounds the difficulty of deriving meaningful medication recommendations. However, a promising approach is to mine transversal information from the EHR [21], [22], i.e., the right side of Fig. 1, enabling the retrieval of relevant insights from other visits, including those of different patients, which can potentially inform more accurate treatment strategies.

Second, the complexity of medical relationships adds another layer of difficulty in formulating reliable medication recommendations [23]. The intricate interdependencies between various medical entities, such as diagnoses, symptoms, and medications, are further complicated by biases present in observable data [24]. These biases hinder the process of learning accurate and representative embeddings of medical information. Moreover, different medications exhibit distinct co-relevance patterns within the EHR [25], making it particularly challenging to model these relationships in a way that ensures optimal drug combinations. Effective extraction and representation of these multifaceted relationships are essential for accurately matching medications to patient profiles.

Within this context, we propose a Relevance aware Visit Similarity Network (named RaVSNet), a drug recommendation model that leverages relevance knowledge and visit similarity networks for accurate drug recommendations. The model introduces a generic pretraining framework with self-supervision and causal effect inference tasks to explore deep connections between medical information and medications while correcting biases in observable data. To address the underutilization of similarity information, we propose a similarity learning approach designed to extract similar visit records of patients from both longitudinal and transversal dimensions. The Relevance Aware Network models the relationship between medical information and medications to obtain accurate medication embeddings. The final recommendation module matches visit representations with medication embeddings, correcting biases using a causal effect matrix for precise recommendations. In summary, the contributions of this paper can be summarized as:

- We propose a network to perform a fine-grained exploration of similar visit records and internal knowledge, fully leveraging multi-dimensional similarity information and the relevance relationship between medical information and medications to enable precise patient modeling and accurate drug recommendations.
- We propose a general pretraining framework for drug recommendation, to improve the expressive ability of patient embeddings by exploring the deep connections within medical information, which can be seamlessly integrated into most drug recommendation methods and achieve performance improvements.
- We have conducted extensive experiments on the public dataset MIMIC-III and MIMIC-IV. The experimental results demonstrate that the proposed algorithm achieves superior levels of accuracy compared to state-of-the-art methods.

The structure of this paper is arranged as follows. Section II presents a comprehensive overview of prominent approaches in medication combination recommendation and techniques for learning medication representations. In Section III, we formally define the research problem. Section IV elaborates on the proposed framework and its corresponding optimization strategy. Section V showcases a series of experiments, including extensive evaluations and comparisons with cutting-edge baselines. Finally, Section VI summarizes the key findings and offers concluding insights.

II. Related Work

A. Drug Recommendation

Drug recommendation employs artificial intelligence technology and patient medical information to deliver safe and effective medication plans. Currently, these methods are mainly categorized into instance-based methods and longitudinal drug recommendation methods. Instance-based methods emphasize the use of medical information from the patient's current visit. For example, LEAP [15] extracts features from the patient's current medical information to perform multi-instance, multi-label drug recommendation. DMNC [26] adopts a dual-memory neural interaction filtering framework, focusing on the interaction between medication and patients to achieve accurate medication prediction. In contrast to instance-based methods, longitudinal drug recommendation methods utilize the temporal dependencies among clinical events, emphasizing the patient's long-term health status. For example, RETAIN [27] employs

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a two-level attention model, focusing on significant visits and medical information within these visit records. GAMENet [28] is the first to use RNN to model patient representations and models medication co-occurrence relationships from EHR data using graph neural networks for medication representation. MICRON [29] uses residual networks to predict changes in patient health status, focusing on the variations between consecutive visits. 4SDrug [30] efficiently models symptom and drug sets through a set-oriented representation and similarity measurement method. COGNet [31] encodes medical and medication information through Transformers and designs a drug "copy or predict" mechanism to incorporate drugs still applicable from the previous visit into the current recommendation. REFINE [32] extracts trend information from multiple visits regarding drug dosage titration and laboratory test reactions, building models to obtain effective patient representations. StratMed [18] adopts a hierarchical strategy to divide data into different levels, strengthens learning on low-frequency data through reinforcement learning, enhances the expression ability of sparse data. Due to the superiority of RNNs in short sequence modeling, a considerable portion of current longitudinal drug recommendation works [22], [33], [34], [35], [36], [37], [38] still follows the visit representation modeling approach of GAMENet. However, the aforementioned approach still has the following limitations: 1. Without filtering longitudinal visit records, as mentioned in the introduction, the patient's current visit condition does not always align with their historical visit records. 2. The lack of modeling for transversal visit similarities reduces recommendation performance, especially for patients with no longitudinal visit history.

B. Medication Representation Learning

Medication representation learning remains an important topic due to the association between medication and the patient's visit status. Carmen [39] obtains medication representation by mining co-occurrence relationships between medications. SHAPE [40] increases drug safety by modelling DDIs to obtain medication representation. Previous studies recognized the medication molecule structure contains important medication properties, such as the efficacy in curing disease. Therefore, SafeDrug [41] further optimizes medication representation through molecular modeling. MoleRec [17] enhances the utilization of molecular structures, emphasizing the importance of molecular substructures in medication molecular representation learning, and achieves higher accuracy in drug recommendation combinations. CSEDrug [42] designs a knowledge-guided triplet learning loss to incorporate multiple types of medication knowledge, including DDIs and molecule structures of drugs. However, medications with similar molecular structures do not always exhibit similar functionality [43]. Additionally, existing methods insufficiently explore the association between medical information and medication in medication representation learning. Only considering drug molecular structures or modeling DDIs relationship knowledge extracted from EHR makes it difficult to clearly match the relationship between medical information and medication, thus constraining the safety and accuracy of drug recommendations.

III. Problem Formulation

In the context of intelligent healthcare, accurate drug recommendation hinges on the effective modeling of patient information derived from electronic health records. Each patient's medical history is composed of a series of visits, which can be further divided into two types of similar visit records. To enable precise and interpretable modeling, it is essential to formally define the structure and components of the input data, the types of similar visit records leveraged for representation learning, and the overall objective of drug prediction. In the following sections, we define the formal representation of EHR data, specify how similar visits are identified and categorized, and present the drug recommendation task as a multi-label prediction problem based on structured clinical information.

Electronic Health Records (EHR): Patient EHR data encapsulates medical information in the form of longitudinal vectors of medical codes. In this study, the medical information considered includes diagnoses:$c_t^d \in \{0,1\}^{|D|}$, procedures:$c_t^p \in \{0,1\}^{|P|}$, symptoms: $c_t^s \in \{0,1\}^{|S|}$ and medications:$c_t^m \in \{0,1\}^{|M|}$. We assume that diagnosis, procedure, symptom, and medication codes are uniformly available for each visit. Each patient's visit is represented as $v_t = [c_t^d, c_t^p, c_t^s, c_t^m]$, where D, P, S and M represent the sets of diagnoses, procedures, symptoms and medications, respectively. The EHR dataset is defined as $V_{EHR} = \{v_1^1, v_2^1, \ldots, v_{t1}^1, v_1^2, v_2^2, \ldots, v_{t_n}^n\}$, where the superscript is the patient number and the subscript is the visit number.

Similar Visit Records: Similar visit records are categorized into longitudinal and transversal types. Longitudinal similar visits are derived from the historical visits of the current patient, while transversal similar visits are obtained from the records of other patients in the EHR dataset. A patient i with t visits is denoted as $Patient_i = [v_1^i, v_2^i, \ldots, v_t^i]$, for the j-th visit, the set of previous visit records is given by $\{v_1^i, v_2^i, \ldots, v_j^i - 1\} \in V_{history}^i$. The problem of identifying longitudinal similar visits involves selecting relevant records $V_{related}^i$ from $V_{history}^i$. For transversal similar visits, the task is to retrieve the Top-K most similar visit records to patient's current visit from V_{EHR}.

Drug Recommendation: The definition of drug recommendation is to learn a function $f(\cdot)$, using all medical information (excluding c_t^m) obtained from a visit, denoted as $[c_1^d, c_2^d, \ldots, c_t^d]$, $[c_1^p, c_2^p, \ldots, c_t^p]$, $[c_1^s, c_2^s, \ldots, c_t^s]$ and $[c_1^m, c_2^m, \ldots, c_{t-1}^m]$ to predict a drug combination $\hat{c}_t^m \in \{0,1\}^{|M|}$:

$$\hat{m}(t) = f\left([c_i^d]_{i=1}^t, [c_i^p]_{i=1}^t, [c_i^s]_{i=1}^t, [c_i^m]_{i=1}^{t-1}\right) \quad (1)$$

The goal of prediction is to make the predicted value $\hat{c}_t^m$ as close as possible to the actual value c_t^m.

IV. Model

In this section, we detail our proposed model and its key components, as shown in Fig. 2. RaVSNet integrates longitudinal and transversal similarity information along with medical relevance knowledge to enhance accuracy and safety. The model comprises three primary modules: the **Visit Similarity Network**, which aggregates longitudinal and transversal similarity information to extract features from patients' current and related visit records; the **Relevance Aware Network**, which models

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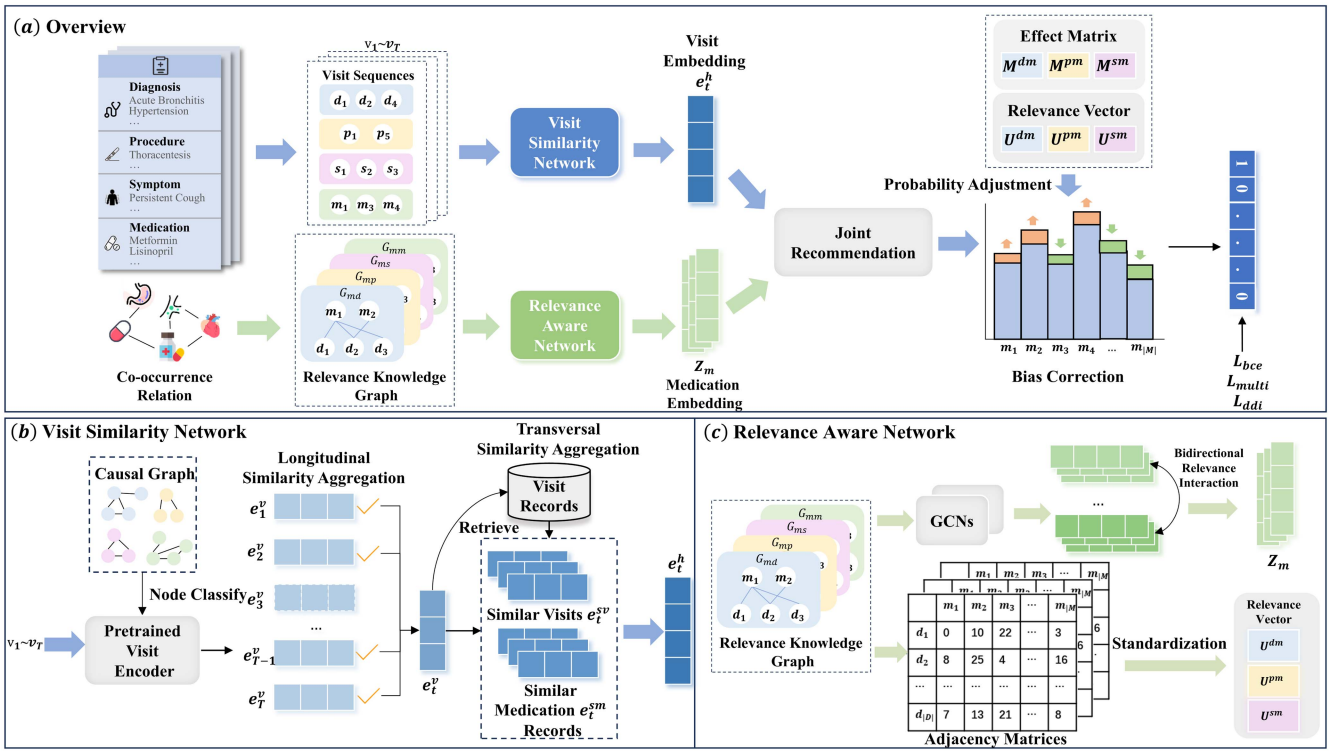


Fig. 2. (a) Overview of RaVSNet (b) Visit Similarity Network, which generates the visit representation using the similarity information; (c) Relevance Aware Network, which is used to mine the internal relevance knowledge in the medical information.

the matching relationships between medical information and medications using relevance knowledge graphs to learn robust medication representations; and the **Joint Recommendation with Bias Correction Module**, which integrates visit and medication representations to generate recommendations while correcting potential biases in the results using causal effect matrices. Furthermore, we propose a generalized pretraining framework for drug recommendation, which captures deep connections between medical information and medications through tasks such as medication sequence reconstruction and causal effect inference, providing accurate initial representations.

A. Visit Similarity Network

1) Visit Representation Generation: This subsection highlights the varying significance of clinical codes in prescribing and introduces a visit similarity network to extract the multi-dimensional similarity information in EHR data, thereby enhancing visit representation accuracy.

First, based on the pretraining model, that will be described in detail in Section IV.D, the visit medical information $v_1 \sim v_T$ is projected into the embedding space, where T is the total number of visits for this patient. Taking the t-th visit as an example, the encoding process yields the representations e_t^d, e_t^p, e_t^s and e_{t-1}^m. Given the varying importance of entities within a session, we propose an adaptive entity classification method based on causal graphs. The causal graphs are also derived from the subsequent pretraining framework in Section IV.D. This method categorizes medical entities into four distinct types: (1) cause nodes (no in-degree, with out-degree); (2) intermediate nodes (with both in-degree and out-degree); (3) effect nodes (with in-degree, no out-degree); and (4) independent nodes (neither in-degree nor out-degree). This classification reflects the significance of clinical codes, and an attention mechanism updates each entity's representation to enhance the expressiveness of key codes. For example, the specific formulation for diagnosis information is as follows:

$$w_t^k = \frac{\exp(W \cdot e_t^{D_k} + b)}{\sum_{j=1}^{4} \exp(W \cdot e_t^{D_j} + b)} \tag{2}$$

$$e_t^D = \sum_{i=1}^{|D_t|} e_t^{d_i} \cdot w_t^k \tag{3}$$

where k represents the k-th entity type and D_k represents the diagnosis set of this type; W and b are the trainable weight matrix and bias; $e_t^{D_k}$ is the sum of all diagnosis embeddings in the D_k. Then updating the diagnosis embedding in the t-th visit by (3). Other medical information can be obtained in the same way, ultimately resulting in the procedure embedding e_t^P, the symptom embedding e_t^S and the historical medication embedding e_{t-1}^M.

Next, the fusion weights of the pretraining model are fine-tuned to integrate clinical information, producing the visit representation e_t^v. This process is applied identically to all other visits, ultimately resulting in $e_1^v \sim e_T^v$.

2) Longitudinal Similarity Aggregation: To remove noisy data from the patient's visit history, this paper proposes a method for

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selecting longitudinally similar visit records, aiming to capture the correlation between current and historical visits comprehensively and filter out historical visit records similar to the current visit condition. This approach enables more effective modeling of the patient's historical visit data. First, we design a correlation-based visit selection module to compute a correlation score between the historical and current visits. Specifically, the current and historical visits are concatenated, and then a nonlinear multi-layer perceptron (MLP) is used to obtain a correlation score for each historical visit, as shown in the following equation:

$$r_t^i = sigmoid(concat(e_t^h, e_t^v)W_s + b_s) \tag{4}$$

where $e_t^h \in V_{history}^i$ represents all historical visits of patient i, and the relevance score r^t of the historical visit is obtained using a sigmoid function.

To ensure the smooth flow of gradients during model training, the set of historically related visits $V_{related}^i$ for patient i is selected based on the relevance scores. The Gumbel-Softmax mechanism is employed to convert the discrete sampling problem into a differentiable continuous problem, as shown in the following equation:

$$y_i^t = \frac{\exp\left(\left(\log(\pi_i) + g_i\right)/\tau\right)}{\sum_{j=1}^{2}\exp\left(\left(\log(\pi_j) + g_j\right)/\tau\right)} \tag{5}$$

$$V_{\text{related}}^i = \left\{v^t : t \in \{1, \ldots, (T-1)\} \text{ and } y_1^t \geq 0.5\right\} \tag{6}$$

where the input $\pi \in \mathbb{R}^{2\times 1}$ to the Gumbel-Softmax is obtained by concatenating the probability of a historical visit being selected, r_t^i, and the probability of not being selected, $1 - r_t^i$; the variable g_i represents a random variable sampled from the Gumbel distribution; τ is the temperature coefficient, which is a hyperparameter. When τ is large, the output of the Gumbel-Softmax becomes smoother, approximating a random distribution. Conversely, as τ decreases, the Gumbel-Softmax output becomes sharper, approaching hard selection, where the item with the highest probability is directly chosen.

After obtaining the related historical visits for patient i, instead of aggregating historical visits sequentially using GRU as in existing methods, we employ an attention mechanism to capture the relevance scores between the current visit and the selected historical visits. These relevance scores are then used to aggregate the historical visits with the current visit, enabling a more comprehensive utilization of the historical visit information. The specific formula is as follows:

$$\alpha^t = \frac{\exp\left(\left(e_t^v W_\alpha e_t^{k^T}\right)/\sqrt{dim}\right)}{\sum_{e_t^k \in V_{\text{related}}^i \cup e_t^v}\exp\left(\left(e_t^v W_\alpha e_t^k\right)/\sqrt{dim}\right)} \tag{7}$$

$$e_t^v = \sum_{e_t^k \in V_{\text{select}}^i \cup v_T} \alpha^t e_t^k \tag{8}$$

where α^t is the attention score, and the patients' current visit representations e_t^v are updated based on the attention score.

3) Transversal Similarity Aggregation: Mining other visits from the EHR is crucial for improving patient care and clinical decision support, particularly in tasks like drug recommendation. Effectively utilizing EHR data enables the discovery of visits that are highly relevant to the patient's current health status, offering valuable references for personalized treatment and decision-making. Therefore, we retrieve the Top-K visits with the highest cosine similarity to the current visit from the visit-level embedding matrix M_v. The similarity calculation formula is shown as follows:

$$s_{t,k} = \frac{e_t^v \cdot e_k^v}{|e_t^v| \cdot |e_k^v|}(e_k^v \in M_v, k \neq t) \tag{9}$$

where $M_v \in \mathbb{R}^{V\times 4*dim}$ consists of all visit representations in the EHR data; e_k^v is the row vector of M_v which represents the other visits.

Then summing the Top-K similar visits' embeddings to obtain $\hat{e}_t^{sv}$. Utilizing the multi-head cross-attention mechanism, e_t^v serves as the query, while $\hat{e}_t^{sv}$ serves as the key and value, resulting in the representation of similar visits, denoted as e_t^{sv}.

Due to the potential absence of historical visit records, existing models face challenges when patients visit for the first time. We believe that medication records of similar visits can help alleviate the problem in drug recommendation. The medication records of the Top-K similar visits are defined as $c_m^k \in \{0,1\}^{|M|}$:, which are projected into embedding space through a learnable embedding table E_m and get embedding e_{sm}^k.

Using the similarity scores $s_{t,k}$ between current visit and k-th visit as weights, the representation of medication for similar visits is obtained by (10):

$$\hat{e}_t^{sm} = \sum_{k=1}^{K} s_{t,k} \cdot e_{sm}^k \tag{10}$$

Given the success of the proposed layer normalization (LN) architecture [44], parameterized $\hat{e}^{sm}$ by a feedforward network, LN is then applied through residual connections and obtain e_t^{sm}:

Finally, the representation of the current visit e_t^v, the representation of similar visits e_t^{sv}, and the representation of medication for similar visits e_t^{sm} are concatenated to obtain the current visit state representation e_t^h.

B. Relevance Aware Network

In drug recommendation systems, the co-occurrence relationships between medical information—such as diagnoses, procedures, and symptoms—and medications are positively correlated with drug prescriptions. Specifically, when a certain diagnosis, procedure, or symptom frequently co-occurs with a particular medication, this suggests that the medication is commonly used in that medical context. Consequently, the higher the co-occurrence frequency of this medical information, the greater the likelihood that the medication will be recommended. To fully utilize the internal knowledge in EHR data, we introduce a Relevance Aware Network based on relevance knowledge mining, aiming to fully explore the matching relationships between medical information and enhance the accuracy of drug recommendations. The following four types of knowledge graphs are generated based on the co-occurrence relationships between medical information and

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medication: medication-diagnosis graph $G_{md} = \{V_{md}, E_{md}\}$, medication-procedure graph $G_{mp} = \{V_{mp}, E_{mp}\}$, medication-symptom graph $G_{ms} = \{V_{ms}, E_{ms}\}$, medication-medication graph $G_{mm} = \{V_{mm}, E_{mm}\}$, where the node set $V_{md} = M \cup D$, $V_{mp} = M \cup P$, $V_{ms} = M \cup S$ and $V_{mm} = M$; E_{md}, E_{mp}, E_{ms} and E_{mm} are edges composed of known co-occurrence combinations in EHR. An edge exists between two nodes if a co-occurrence relationship is observed between the corresponding clinical codes in the dataset. The adjacency matrices for these graphs are represented as A_{md}, A_{mp}, A_{ms}, and A_{mm}, respectively, capturing the strength of these co-occurrence relationships.

1) Relevance Information Extraction: To further leverage the knowledge of the correlations between medical information and medications, we designed a relevance information extraction module. This module uses medical information as an index to extract the co-occurrence information of medical information and medications relevant to the current visit. The co-occurrence information is extracted from the adjacency matrix A_* of the relevance knowledge graph. For example, the adjacency matrix of the medication-diagnosis relevance knowledge graph G_{md} is represented as A_{md}. The method for extracting co-occurrence information from all relevance knowledge graphs is as follows:

$$U_e[j] = \sum_{i=1}^{|c^*|} A_*[c_t^*, j] \tag{11}$$

Here, the j-th element of $U_e \in \mathbb{R}^{|M|}$ represents the total co-occurrence count between the j-th medication in the EHR data and all medical information associated with the patient. Since this value is positively correlated with the probability of recommending the j-th drug, this vector will be further utilized in generating the final recommendation probabilities in Section D.

2) Relevance Interaction Module: In this module, the Medical Information Relevance Knowledge Graphs are fed into a two-layer GCN [45] to learn embeddings based on various co-occurrence relationships for each graph. Ultimately obtained the medication representation Z_{md}, Z_{mp}, Z_{ms} and Z_{mm} from the medication-diagnosis graph, the medication-procedure graph, the medication-symptom graph, and the medication-medication respectively.

Recognizing the significance of drugs' interactions within a population, rather than in isolation, we introduce a Relevance Interaction Module (RIM) inspired by the Set Attention Block [46] to model their higher-order interactions. The RIM is defined as follows:

$$\begin{aligned} &RIM(X, Y) = LN(Z + FF(Z)) \\ &\text{where} \quad Z = LN(X + Attention(X, Y, Y)) \end{aligned} \tag{12}$$

We apply a bidirectional RIM to exchange information and adjust the representations between each level to learn the following joint representation:

$$\tilde{Z}_{ij} = RIM(Z_i, Z_j) \tag{13}$$

where $Z_i, Z_j \in \{Z_{md}, Z_{mp}, Z_{ms}, Z_{mm}\}$ and $Z_i \neq Z_j$

Then, the self-attention mechanism are applied to establish internal connections at each level, updating the representation of each layer. Finally, by concatenating the representations of each level, the medication representation is obtained, which is rich in relevance knowledge, denoted by Z_m.

C. Joint Recommendation With Bias Correction

The Joint Recommendation with Bias Correction module includes both joint drug recommendation and bias correction based on causal effects. Inspired by ACDNet [47], we integrate visit and medication representations for joint drug recommendation. This process involves two sub-modules: sequence-aware recommendation and similarity-aware recommendation.

A sequence aware recommendation is a nonlinear transformation $\mathcal{F} : \mathbb{R}^{dim} \rightarrow \mathbb{R}^{|M|}$, the drug recommendations o_1 are generated based on the patient's current visit representation e_t^h, aiming to capture the health status in the visit sequence and generate direct recommendation results. A straightforward solution is to implement $\mathcal{F}$ with a neural network, as follows:

$$o_1 = \mathcal{F}(e_t^h; \Theta_1) \tag{14}$$

where $\mathcal{F}$ is a multi-layer perceptron (MLP), and Θ_1 denotes the parameters of the MLP for sequence aware recommendation.

In the similarity aware recommendation, to match the score between current visit status and each medication, we calculate the similarity between the patient's current visit and the medication representation, denoted as $s_m \in \mathbb{R}^{|M|}$. Then, the similarity results and the direct recommendation results are combined as input into a neural network, guiding the recommendation restructuring jointly to generate the combination recommendation results o_2:

$$o_2 = \mathcal{F}(\alpha o_1 \# \beta s_m; \Theta_2) \tag{15}$$

where $\alpha, \beta \in \mathbb{R}^1$ are learnable fusion weights; Θ_2 denotes the parameters of the MLP for similarity aware recommendation; # is the concatenation function.

Finally, leveraging the attention mechanism, the weights of the representations for both sequence aware and similarity aware recommendations are adjusted to derive the joint drug recommendation result. Moreover, to achieve a more accurate probability distribution, we leverage the co-occurrence activation layer extracted by the relevance-aware module. The underlying rationale is that medical information can directly influence clinicians' prescription decisions for discharge medications. Therefore, the co-occurrence information U_e is utilized to directly adjust the probability distribution. The final joint drug recommendation probability is formulated as follows:

$$\hat{o} = \sigma(w_1 \odot o_1 + w_2 \odot o_2 + std(U_e)) \tag{16}$$

where $\hat{o} \in \mathbb{R}^{|M|}$ is the joint drug recommendation result; σ is a non-linear activation function; w_1, w_2 are the learnable weight parameter; std represents the standardization operation on vectors U_e.

To address biases arising from uneven data distribution, this paper introduces a causal inference approach to adjust the recommendation probability for each medication, correcting biases

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caused by embedding matching. Based on causal effect matrices generated during pretraining, which will be described in detail in Section IV.D, we increase the recommendation probability for medications with significant therapeutic effects and decrease it for those below a certain threshold. The specific formulation is as follows:

$$\hat{o}'(m_i) = \begin{cases} \hat{o}(m_i)+\phi 1, & \text{if} \quad M^*_{mi-*} \geq \delta 1 \\ \hat{o}(m_i)-\phi 2, & \text{if} \quad M^*_{mi-*} \leq \delta 2 \end{cases} \tag{17}$$

where $\hat{o}'(m_i)$ is the probability that m_i is recommended; $\phi 1$ and $\phi 2$ are a learnable probability-adjusted values; $\delta 1$ and $\delta 2$ are predefined threshold hyperparameters; M^*_{mi-*} is the quantified value obtained from the pretrained causal effects matrix, for example, $M^{dm}_{mi-D_t}$ represents the value of the causal effect of each diagnosis and m_i in the diagnosis sequence of the t-th visit.

Drug recommendation is framed as a multi-label prediction problem. Therefore, we employed two standard loss functions for training: binary cross-entropy loss (L_{bce}) and multi-label margin loss (L_{multi}), defined as follows:

$$L_{bce} = -\sum_{i=1}^{|M|} m_i \log(\hat{o}_i) + (1-m_i)\log(1-\hat{o}_i) \tag{18}$$

$$L_{multi} = \sum_{i,j:m_i=1,m_j=0} \frac{\max(0,1)-(\hat{o}_i-\hat{o}_j)}{|M|} \tag{19}$$

where m_i and $\hat{o}_i$ are the i-th entries.

Next, we define the DDI loss as the absolute difference between the drug-drug interaction values in the ground truth and the predicted drugs. The formula is as follows:

$$L_{ddi} = \left| \sum_{i=1}^{|M|} \sum_{j=1}^{|M|} a^{ddi}_{ij} \cdot (\hat{m}_i \cdot \hat{m}_j - m_i \cdot m_j) \right| \tag{20}$$

Ultimately, the training loss of the model can be expressed as:

$$L = \beta(\lambda L_{bce} + (1-\lambda)L_{multi}) + (1-\beta)L_{ddi} \tag{21}$$

where λ and β are predefined hyperparameters.

D. Pretraining Framework

The Pretraining Framework incorporates two tasks aimed at improving visit and medication representations, as shown in Fig. 3. These tasks are designed to exploit the inherent patterns and relationships among clinical codes, thereby revealing deeper connections between them. By pretraining on these tasks, we aim to more effectively capture the complexities of various clinical conditions. Moreover, the proposed pretraining framework is designed to be universal, enabling seamless integration with existing drug recommendation models to enhance their performance and adaptability.

a) Task 1: Medication Sequence Reconstruction (MSR).: During clinical visits, physicians make medication recommendations based on the specific circumstances of the patient's current visit. Therefore, this paper introduces a self-supervised visit representation learning method to capture visit-level embeddings from complex medical information. Specifically, the clinical codes are extracted from a single-visit patient record within EHR data. Similar to the Masked Language Model (MLM) task, special tokens [MASK] are randomly used to replace the original clinical codes. Then the codes are projected into an embedding space using four learnable embedding tables:

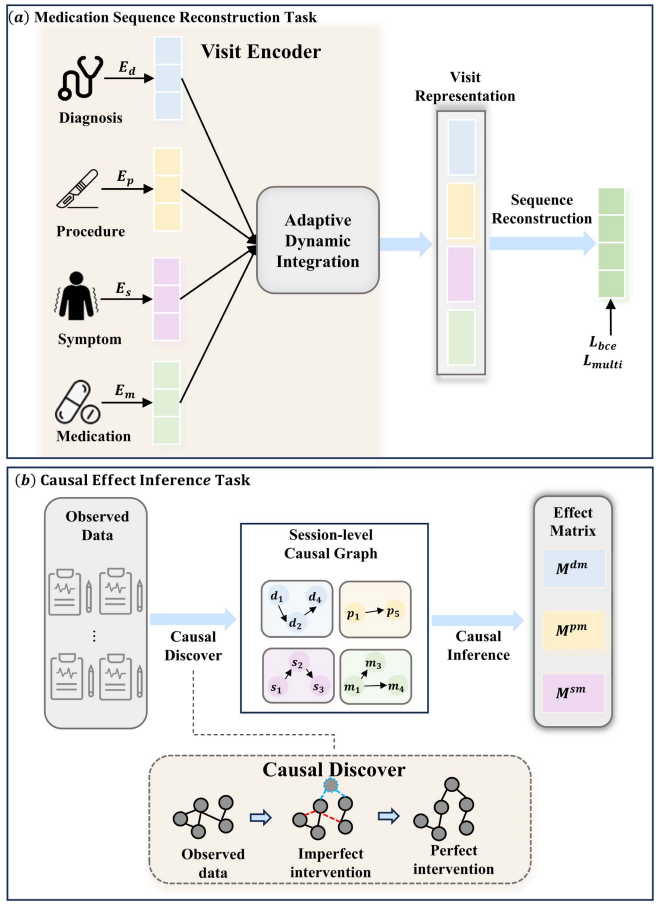


Fig. 3. Pretraining Framework. (a) Medication Sequence Reconstruction task. (b) Causal Effect Inference task.

$$e^*_t = c^*_t E_* \tag{22}$$

where $c^*_t \in \{0,1\}^{|C^*|}$ indicates the unified definition for different types of medical codes, C^* is the specific medical sets; $E_* \in \mathbb{R}^{|C^*| \times \dim}$ represents the embedding tables for these codes with dim as the embedding dimension. These tables are learnable and shared across patient visits.

Unlike existing works that directly concatenate embeddings of medical information, we believe that each type of medical information influences a doctor's prescribing decision differently. Therefore, we propose an adaptive fusion method to dynamically adjust fusion weights, as described in (22):

$$e^v_t = \left(w_1 e^d_t \# w_2 e^p_t \# w_3 e^s_t \# w_4 e^m_t\right) \tag{23}$$

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where $w_1, w_2, w_3, w_4 \in \mathbb{R}^1$ are learnable fusion parameters used to dynamically adjust the weights of different clinical codes. $\#$ represents the concatenation operation.

Next, the embedding results are transformed into the dimension of $|M|$ through a multi-layer perceptron with skip-connections and layer normalization to reconstruct the medication sequence for a specific patient visit. Through this mapping, the embedding vectors derived from medical codes are reconstructed into a representation consistent with the structural format of the medication sequence, expressed by the following formula:

$$e_t = \sigma(LN(e_t^v + e_t^v W_{LN})) \tag{24}$$

where σ is a nonlinear activation function, W_{LN} is a learnable weight parameter and LN is layer normalization.

The output strategy, loss function, and training method of the Pretraining Framework align with those of the main module.

b) Task 2: Causal Effect Inference (CEI).: In recommendation systems, there is a well-known issue of recommendation bias caused by the limitations of observable information. This paper posits that a similar problem exists in the domain of medication recommendation. To address this, we employ causal analysis methods to analyze and quantify the causal relationships between medical information and medications. Specifically, we first generate Bayesian equivalence classes based on the data distribution U in the EHR using an equivalence score criterion $S(G, U)$. The Bayesian equivalence classes are then optimized through the Greedy Intervention Equivalence Search (GIES) algorithm [48], which generates a causal graph from the original data. The specific formulation is as follows:

$$S(G, U) = \sum_{i=1}^{n} S(N_i, P_i^G) \tag{25}$$

where n represents the total number of nodes in the Bayesian network, N_i denotes the i-th node, and P_i^G represents the parent nodes of N_i in the graph G.

GIES is a score-based Bayesian algorithm that heuristically searches for the graph that minimizes a likelihood score on the data. Finally, the isomorphic session-level causal graphs $G_i^d, G_i^p, G_i^s, G_i^m$ are constructed for different medical codes from the i-th visit, encompassing the various influencing factors of clinical codes.

Next, a causal inference algorithm is employed to quantify the impact of different medical information on medications, revealing the deeper connections between medical information and drug prescriptions. Specifically, the causal graph is represented as binary variables and a Generalized Linear Model (GLM) is used to simulate the causal effects between clinical codes and medications. This process results in three causal effect matrices for diagnosis-medication, procedure-medication, and symptom-medication, denoted as $M^{dm} \in \mathbb{R}^{|D|\times|M|}$, $M^{pm} \in \mathbb{R}^{|P|\times|M|}$, and $M^{sm} \in \mathbb{R}^{|S|\times|M|}$. The values in these causal effect matrices represent the likelihood that a medication will respond to a patient with specific medical information, elucidating the causal effects of particular medical factors on drug treatment.

TABLE I
DATA STATISTICS

Items	MIMIC-III	MIMIC-IV
# of patients	6313	54845
# of visits	14599	138824
avg. # of visits	2.31	2.53
max. # of visits	29	69
diag. space size	1957	2000
prod. space size	1416	1500
sym. space size	434	564
med. space size	131	131
total # of DDI pairs	448	448

V. EXPERIMENTS

A. Dataset and Metrics

The model in this study was tested on the publicly available MIMIC-III [49] and MIMIC-IV [50] datasets. To ensure a fair comparison with prior work and to facilitate seamless integration with potential future healthcare datasets, we adopted the data processing methods proposed by previous studies [18], [41]. Data from different sources are independently extracted and merged based on patient and visit identifiers. Diagnosis, procedure and symptom codes are encoded as multi-hot vectors. DDI information is incorporated from the TWOSIDES database [51], focusing on the top 40 most severe interaction types at the ATC third level. The statistical details of the final dataset are shown in Table I:

Referring to previous work [17], [18], [41], we evaluated the model for drug recommendation using three popular metrics: Precision Recall AUC (PRAUC), Jaccard Similarity Score (Jaccard), and F1-Score. Previous studies generally assume that a lower Drug-Drug Interactions (DDI) score is better. However, DDIs are also present in real-world prescriptions by physicians. For instance, the ground truth DDI score in the MIMIC-III dataset is 0.0904, while in the MIMIC-IV dataset, it is 0.0883. Similarly, studies [31] and [52] have pointed out that a lower DDI score is not always preferable. To address this, we propose the ΔDDI metric to quantify the discrepancy between the predicted DDI score and the ground truth DDI score.

The metrics details are as follows:

The PR_AUC value for patient j at the t-th visit is calculated by:

$$PR_AUC = \sum_{k=1}^{|M|} P_k \left(R_k - R_{(k-1)}\right) \tag{26}$$

$$R = \frac{|\{i : m_{j,i} = 1\} \cap \{i : \hat{m}_{j,i=1}\}|}{|\{i : m_{j,i} = 1\}|} \tag{27}$$

$$P = \frac{|\{i : m_{j,i} = 1\} \cap \{i : \hat{m}_{j,i=1}\}|}{|\{i : \hat{m}_{j,i} = 1\}|} \tag{28}$$

where m_j is the ground-truth drug combination during the t-th visit of patient j, and $m_{j,i}$ is the i-th element; $|\cdot|$ denotes the cardinality; $\cap$ is set interaction; k is the rank in the sequence of the retrieved drugs; P_k and R_k is the precision and recall at cut-off k in the ordered retrieval list.

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The Jaccard coefficient for patient j at the t-th visit is calculated by:

$$Jaccard = \frac{|\{i : m_{j,i} = 1\} \cap \{i : \hat{m}_{j,i=1}\}|}{|\{i : m_{j,i} = 1\}| \cup |\{i : \hat{m}_{j,i} = 1\}|} \tag{29}$$

The F1 score is the harmonic mean of precision and recall, are calculated by:

$$F1 = \frac{2R \times P}{R + P} \tag{30}$$

ΔDDI rate for patient j is calculated as below:

$$DDI = \frac{\sum_{l,k \in \{i:\hat{m}_i=1\}} A_{l,k}}{\sum_{l,k \in \{i:\hat{m}_i=1\}} 1} \tag{31}$$

$$\Delta DDI = |DDI_{gt} - DDI_{predict}| \tag{32}$$

where A_{lk} represents the drug interaction matrix and its corresponding position is 1 if there is interaction between two drugs and 0 otherwise. DDI_{gt} is the ground truth DDI score of patient j, and $DDI_{predict}$ is the drug recommendation results.

B. Performance Comparison

We compared RaVSNet with several baselines, which can be categorized into machine learning-based methods and deep learning-based approaches. Two traditional machine learning methods and thirteen state-of-the-art deep learning-based method were selected as baselines:

- Logistic Regression (LR) serves as an instance-based classifier with L2 regularization.
- Ensemble Classifier Chain (ECC) [53] is a widely used approach for multi-label classification.
- RETAIN (NIPS'16) [27] utilizes a two-layer RNNs with reverse-time attention to capture key insights from visits and clinical codes.
- LEAP (KDD'17) [15] models drug recommendation as a sequential task, treating each recommendation as a step in generating a sentence.
- DMNC (KDD'18) [26] uses a Memory-Augmented Neural Network to construct drug recommendations.
- GAMENet (AAAI'19) [28] integrates internal knowledge from EHR data and capturing information about internal knowledge using graph neural networks and memory mechanisms.
- SafeDrug (IJCAI'21) [41] provides drug recommendations by modelling drug molecules.
- MICRON (IJCAI'21) [29] updates their historical medication combinations when they are new to the clinic.
- COGNet (WWW'22) [31] introduces a translation model that generates drug recommendations through a 'copy-or-predict' mechanism and uses a transformer decoder.
- MoleRec (WWW'23) [17] highlights the importance of molecular substructures for drug recommendation.
- StratMed (KBS'24) [18] optimises drug combinations by coordinating the safety and accuracy of drugs through a bi-graph network structure.
- DAI-Net (JBHI'24) [39] uses the patient's longitudinal representation as a query to decode the matching information between symptoms and drug molecules.
- VITA (AAAI'24) [20] selects the K most similar historical visits to the patient's current visit from their medical records.
- RAREMed (SIGIR'24) [24] integrates pretraining techniques to capture rare disease patient phenotypes.
- CausalMed (CIKM'24) [25] model the causal relationships between medications and specific health state of patients.

The comparison results are shown in Table II.

Table II shows that traditional machine learning methods like LR and ECC, while applicable to drug recommendation, perform poorly across multiple metrics, underscoring the shift towards deep learning for improved accuracy and safety in drug recommendation.

Among deep learning methods, instance-based models like LEAP perform the worst, due to their inability to capture temporal progression or relational dependencies across visits. Although DMNC introduces memory-augmented networks, it still underperforms due to inadequate modeling of visit-level medical relevance and drug co-prescription logic. In contrast, longitudinal-based methods like GAMENet, SafeDrug, MICRON, and MoleRec leverage patients' historical visits, showing improved prediction accuracy. However, they fail to fully capture the complex relationships between medications and medical information, which limits their accuracy. COGNet incorporates a copy-predict mechanism to reuse medications across visits, which improves performance but relies heavily on historical prescriptions, making it inapplicable for patients with only a single visit. StratMed uses reinforcement learning on low-frequency data but does not explore inter-visit or inter-patient relationships between patient visits, limiting its performance due to inadequate patient representation learning. DAI-Net emphasizes coarse-grained interactions between different symptom types and fine-grained interactions between drug molecules and symptom types. However, its insufficient exploration of transversal or longitudinal visit records results in suboptimal predictive accuracy. While VITA filters longitudinal visit records to some extent, effectively reducing noise, it similarly lacks a comprehensive exploration of transversal visit records, which limits its performance on sparse datasets. RAREMed integrates pretraining techniques for rare disease representation, but its simplistic visit encoder structure fails to align medical and medication embeddings, restricting the model's ability to fully capture visit representations and ultimately limiting recommendation accuracy. CausalMed introduces a causality-based patient modeling framework that explicitly estimates point-to-point relationships between diseases, procedures, and medications based on patient-specific health states. While this design allows for more personalized entity representations and achieves the highest accuracy among baselines, its overall performance still lags behind RaVSNet. CausalMed focuses primarily on static visit-level causal inference, lacking a dynamic selection mechanism to distinguish clinically relevant from irrelevant visit information.

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TABLE II
PERFORMANCE COMPARISON ON MIMIC-III AND MIMIC-IV

Model	MIMIC-III				MIMIC-IV			
	PR_AUC↑	Jaccard↑	F1-Score↑	$\Delta DDI\downarrow$	PR_AUC↑	Jaccard↑	F1-Score↑	$\Delta DDI\downarrow$
LR	0.7548 ± 0.0031	0.4924 ± 0.0027	0.6496 ± 0.0025	0.0413 ± 0.0007	0.6613 ± 0.0022	0.4366 ± 0.0020	0.6064 ± 0.0025	0.0397 ± 0.0005
ECC	0.7590 ± 0.0026	0.4856 ± 0.0031	0.6438 ± 0.0028	0.0388 ± 0.0007	0.6530 ± 0.0019	0.4223 ± 0.0025	0.6129 ± 0.0018	0.0409 ± 0.0007
RETAIN	0.7600 ± 0.0035	0.4871 ± 0.0038	0.6473 ± 0.0033	0.0379 ± 0.0012	0.6801 ± 0.0025	0.4133 ± 0.0022	0.5785 ± 0.0021	0.0344 ± 0.0004
LEAP	0.6555 ± 0.0035	0.4526 ± 0.0035	0.6147 ± 0.0036	0.0426 ± 0.0006	0.6059 ± 0.0025	0.4145 ± 0.0020	0.5794 ± 0.0023	0.0410 ± 0.0005
DMNC	0.6731 ± 0.0024	0.4563 ± 0.0031	0.6168 ± 0.0032	0.0395 ± 0.0010	0.6138 ± 0.0019	0.4187 ± 0.0025	0.5771 ± 0.0021	0.0352 ± 0.0003
GAMENet	0.7656 ± 0.0046	0.4994 ± 0.0033	0.6560 ± 0.0031	0.0332 ± 0.0005	0.6651 ± 0.0014	0.4446 ± 0.0017	0.5944 ± 0.0027	0.0328 ± 0.0003
SafeDrug	0.7660 ± 0.0022	0.5126 ± 0.0027	0.6692 ± 0.0025	0.0658 ± 0.0003	0.6767 ± 0.0024	0.4470 ± 0.0019	0.6041 ± 0.0025	0.0593 ± 0.0006
MICRON	0.7489 ± 0.0034	0.5219 ± 0.0021	0.6761 ± 0.0018	0.0347 ± 0.0002	0.6739 ± 0.0015	0.4521 ± 0.0023	0.6007 ± 0.0019	0.0372 ± 0.0002
COGNet	0.7707 ± 0.0021	0.5316 ± 0.0020	0.6644 ± 0.0018	0.0318 ± 0.0004	0.6524 ± 0.0025	0.4616 ± 0.0024	0.6172 ± 0.0023	0.0328 ± 0.0003
MoleRec	0.7748 ± 0.0022	0.5301 ± 0.0025	0.6841 ± 0.0022	0.0320 ± 0.0009	0.7002 ± 0.0021	0.4625 ± 0.0018	0.6180 ± 0.0019	0.0324 ± 0.0005
StratMed	0.7779 ± 0.0024	0.5321 ± 0.0027	0.6861 ± 0.0023	0.0330 ± 0.0003	0.7023 ± 0.0014	0.4560 ± 0.0007	0.6122 ± 0.0007	0.0319 ± 0.0002
DAI-Net	0.7717 ± 0.0026	0.5253 ± 0.0030	0.6798 ± 0.0029	0.0329 ± 0.0003	0.7035 ± 0.0018	0.4552 ± 0.0013	0.6114 ± 0.0006	0.0315 ± 0.0001
VITA	0.7635 ± 0.0029	0.5261 ± 0.0033	0.6785 ± 0.0027	0.0388 ± 0.0004	0.6989 ± 0.0016	0.4816 ± 0.0028	0.6200 ± 0.0026	0.0376 ± 0.0003
RAREMed	0.7811 ± 0.0034	0.5360 ± 0.0024	0.6853 ± 0.0030	0.0354 ± 0.0007	0.7295 ± 0.0021	0.4751 ± 0.0017	0.6182 ± 0.0015	0.0347 ± 0.0006
CausalMed	0.7831 ± 0.0023	0.5367 ± 0.0018	0.6898 ± 0.0018	0.0323 ± 0.0007	0.7236 ± 0.0017	0.4757 ± 0.0010	0.6258 ± 0.0010	0.0501 ± 0.0004
RaVSNet	**0.8138** ± 0.0022	**0.5733** ± 0.0021	**0.7208** ± 0.0019	**0.0297** ± 0.0003	**0.7471** ± 0.0017	**0.4970** ± 0.0012	**0.6497** ± 0.0013	**0.0288** ± 0.0001

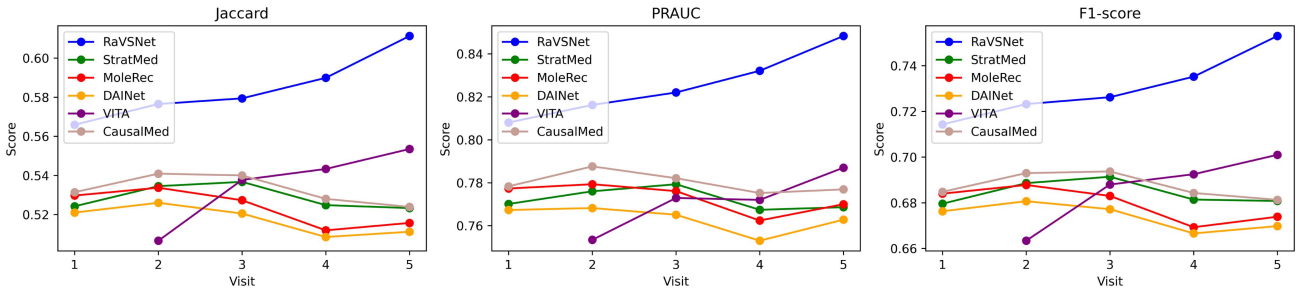


Fig. 4. Performance of different models across multiple visits.

RaVSNet achieves superior performance over a wide range of baseline models due to three key architectural innovations. First, unlike methods that rely solely on longitudinal information (e.g., GAMENet, MoleRec, DAI-Net), RaVSNet explicitly models both longitudinal and transversal similar visits. This dual-perspective similarity network enables the retrieval of clinically relevant context even for patients with few or no prior visits, enhancing the robustness of the model in both sparse and dense EHR scenarios. Second, RaVSNet introduces a Relevance-aware Network to model fine-grained matching relationships between medical concepts and medications. This module captures latent co-dependencies and structured associations that prior models largely overlook, which rely primarily on disease-medication or medication-medication reasoning. Third, the general pretraining framework incorporating MSR and CEI enhances the model's ability to encode temporal dynamics and correct for confounding factors in patient data. These pretraining objectives provide a stronger initialization that improves both convergence and downstream performance, as confirmed by ablation studies.

In order to provide a more detailed explanation of the reasons for our model's advantages, we compared the performance of different models under different numbers of visits. Note that the input unit of the RAREMed model is a single visit rather than the entire patient record. Consequently, the model focuses solely on analyzing the current visit without leveraging historical information and is therefore excluded from the comparison. The experimental results are shown in Fig. 4.

Fig. 4 shows the performance of different models across multiple visits. Among the compared algorithms, StratMed, MoleRec, DAINet, and CausalMed do not perform any visit filtering and are prone to noise when visit histories are long. VITA partially mitigates this by filtering longitudinal visits, but it still underperforms when historical data is sparse or unavailable. In contrast, RaVSNet consistently achieves superior performance across all visit frequencies. This can be attributed to the Visit Similarity Network, which dynamically retrieves relevant information from both longitudinal and transversal visit records. This component enable RaVSNet to robustly identify and leverage clinically meaningful patterns regardless of the density or sparsity of a patient's medical history. As the number of visits increases, RaVSNet's advantage becomes more pronounced, demonstrating its scalability and effectiveness in handling complex patient timelines. Importantly, even in the low-visit regime where many baseline methods struggle due to limited historical context, RaVSNet maintains strong performance by retrieving informative similar visits from other patients. This result confirms the model's adaptability and generalization capability in both short-term and long-term care scenarios.

C. Baselines With Proposed Pretraining Framework

In section IV.D, we claimed that our pretraining framework can be seamlessly integrated into most drug recommendation systems. To validate this claim, we compare the Jaccard score of five baselines (SafeDrug, MoleRec, DAINet, RAREMed and

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TABLE III
PERFORMANCE COMPARISON OF ABLATION STUDY

Model	MIMIC-III				MIMIC-IV			
	PR_AUC↑	Jaccard↑	F1-Score↑	$\Delta DDI\downarrow$	PR_AUC↑	Jaccard↑	F1-Score↑	$\Delta DDI\downarrow$
RaVSNet	**0.8138**	**0.5733**	**0.7208**	**0.0297**	**0.7471**	**0.4970**	**0.6497**	**0.0288**
w/o MSR	0.8034	0.5655	0.7043	0.0301	0.7383	0.4914	0.6444	0.0293
w/o CEI	0.8067	0.5694	0.7072	0.0299	0.7395	0.4932	0.6457	0.0292
w/o Pre	0.7985	0.5562	0.6917	0.0307	0.7371	0.4911	0.6411	0.0301
w/o VSN	0.8016	0.5583	0.6989	0.0311	0.7335	0.4882	0.6387	0.0304
w/o RAN	0.8083	0.5677	0.7081	0.0300	0.7421	0.4933	0.6427	0.0293

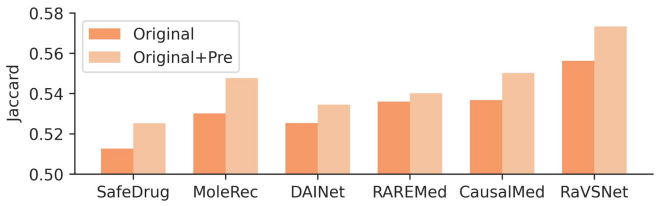


Fig. 5. Accuracy comparison of baselines and their variants with pretraining framework of RaVSNet.

CausalMed) and their variants, each of which is equipped with the proposed pretraining framework. Therefore, the baseline models replace the visit encoders with the pretrained encoder and incorporate a bias correction module based on causal effects. The results of the methods are shown in Fig. 5.

As shown in Fig. 5, the variants of existing methods equipped with our pretraining framework outperform the original methods in Jaccard score. Traditional drug recommendation models rely heavily on patient visit histories. However, these models often fail to capture deeper, latent relationships between medical events and drug prescriptions. The proposed pretraining framework, particularly through MSR, allows the model to learn sequential patterns in medication administration. By incorporating CEI, the model can explicitly model the causal relationships between diseases and medications, allowing the system to better distinguish correlation from causation. By introducing a general pretraining framework with MSR and CEI, RaVSNet effectively improves baseline models by generating more comprehensive patient embeddings, providing a richer and more nuanced understanding of patient medical histories. This verifies that our proposed pretraining framework can improve the accuracy of existing methods.

D. Ablation Study

To evaluate the effectiveness and necessity of each component of RaVSNet, the model is deconstructed by removing these components to create variants as follows: w/o MSR: The MSR task is removed, resulting in the random initialization of the visit encoder parameters. w/o CEI: The CEI task is removed, resulting in failure of adaptive node classification and bias correction. w/o Pretrain: Both pretraining tasks are removed. w/o VSN: The Visit Similarity Network is removed, no longer considering the utilization of similar visit information in EHR data. w/o RAN: The Relevance Aware Network is removed, and using only the visit representation for drug combination prediction.

Table III shows the effects of different variant models of RaVSNet, and the results indicate that each module improves prediction accuracy, which aligns with our expectations. The results of removing one or two pretraining tasks indicate that pretraining plays a crucial role in influencing the outcomes. By defining the medication sequence reconstruction task, the model's representation capability is enhanced. Furthermore, the causal effect inference tasks lay a solid foundation for entity node classification and recommendation debiasing in subsequent modules, significantly boosting prediction accuracy. The w/o VSN results suggest that mining similar visit records enables the use of these records as critical auxiliary information, supplementing and enriching the current visit representation to improve prediction accuracy. The w/o RAN results show that by exploring the relevance knowledge between medical information and medications, and modeling higher-order medication interactions using the relavance aware network, the matching relationship between medication and medical information during drug representation modeling is clarified, leading to more accurate medication recommendations.

E. Parameter Sensitivity Analysis

The longitudinal visit similarity network is one of the important contributions of this paper, and the setting of the hyperparameter τ directly affects the performance of the model.When τ approaches 0, the selection of longitudinally similar visits behaves similarly to the argmax operation, forcing the model to select the historical visit most similar to the current one. As τ increases, the selection of longitudinally similar visits gradually approaches a random distribution. We examined the impact of different values of τ, and the results are shown in Fig. 6.

On the MIMIC-III dataset, $\tau = 0.4$ is identified as the optimal point, while on the MIMIC-IV dataset, the optimal value is $\tau = 0.5$. These settings better capture the underlying similarities between longitudinal visits, thereby improving performance metrics. When τ is too low, the model tends to select the historical visit most similar to the current one, which can lead to overfitting to specific historical visits. Conversely, when τ is too high, the model's selection becomes increasingly random, introducing noise and degrading performance. This highlights that a balanced approach to historical visit selection significantly enhances the model's overall performance.

Another important hyperparameter is K. The value of K controls the dimensionality of transversal similar visits. A smaller K value results in fewer selected transversal similar visit records,

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TABLE IV
IMPACT OF K VALUES ON MIMIC-III AND MIMIC-IV DATASET

Top-K	MIMIC-III				MIMIC-IV			
	PR_AUC↑	Jaccard↑	F1-Score↑	$\Delta DDI\downarrow$	PR_AUC↑	Jaccard↑	F1-Score↑	$\Delta DDI\downarrow$
1	0.8064	0.5645	0.7148	0.0301	0.7378	0.4907	0.6408	0.0300
3	0.8089	0.5685	0.7165	0.0300	0.7392	0.4915	0.6425	0.0294
6	0.8103	0.5699	0.7180	0.0298	0.7409	0.4933	0.6457	0.0290
10	0.8121	0.5722	0.7185	0.0300	0.7421	0.4961	0.6463	**0.0286**
15	**0.8138**	0.5733	**0.7208**	**0.0297**	0.7449	**0.4973**	0.6479	0.0292
20	0.8126	**0.5735**	0.7203	0.0302	**0.7471**	0.4970	**0.6497**	0.0288
25	0.8117	0.5698	0.7173	0.0303	0.7466	0.4955	0.6481	0.0295
30	0.8086	0.5652	0.7131	0.0307	0.7451	0.4936	0.6459	0.0293

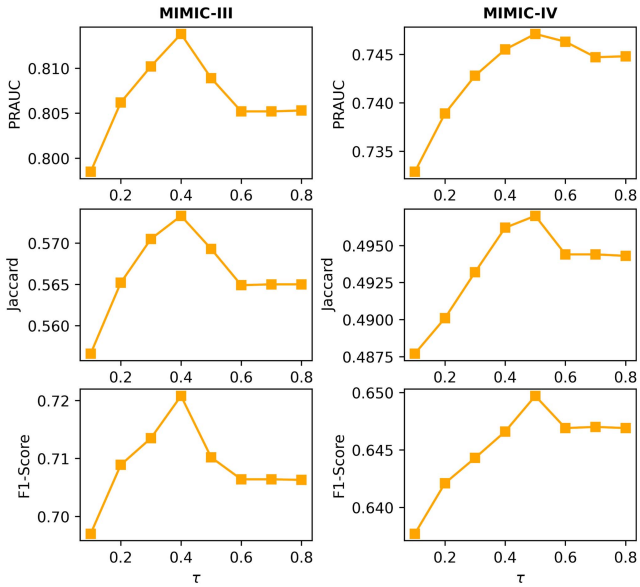


Fig. 6. Impact of τ values on MIMIC-III and MIMIC-IV Dataset.

while a larger K value leads to more selected transversal similar visit records. We examined the impact of different values of K, and the results are shown in Table IV.

Table IV shows that setting the value of K too small or too large reduces the model's performance, highlighting the importance of appropriate hyperparameter selection. A small K overlooks relevant similar visit information in the EHR data, leading to uboptimal use of information. Conversely, a large K introduces noise by allowing dissimilar visits to affect drug recommendations and increases running time, reducing model efficiency.

F. Case Study

We randomly selected a patient from the test set with 5 visits for case analysis. By running both baseline models and our algorithm, we predicted the drug combinations needed for each of the patient's visits to forecast the treatment outcomes. To facilitate the comparison, a concise and visual representation is shown in Fig. 7. In the figure, "Correct" indicates the number of correctly predicted drugs, "Missed" represents the drugs present in the labels but not predicted by the model, and "Unseen" refers to drugs not present in the labels but predicted by the model.

In this case, the patient exhibits a complex condition requiring multiple drug combinations, with treatment plans dynamically adjusted over time. This reflects the personalized and intricate nature of disease management. Among the models evaluated, RaVSNet demonstrates the best performance at each visit, effectively predicting the correct medications while minimizing incorrect predictions, making it well-suited for clinical decision support. Across the five visits, RaVSNet accurately identifies the majority of true medications, achieving the highest Correct values (e.g., 24 correct predictions in the first and the fifth visit). Simultaneously, its low Unseen values indicate that the predicted medications closely align with actual clinical needs. The model's superior performance can be attributed to its unique visit similarity network architecture, which effectively captures complex inter-visit medication relationships and enables precise predictive capabilities. Based on the model's performance, RaVSNet excels at capturing both long-term maintenance medications and dynamically adjusted medications. This makes it particularly suitable for complex treatment scenarios involving polypharmacy, effectively assisting clinicians in developing personalized treatment plans. VITA, which incorporates filtering of longitudinal visit records, shows stable performance in predicting maintenance therapies, making it well-suited for managing chronic diseases. However, it is slightly inferior to RaVSNet in terms of Unseen values, with some erroneous predictions potentially unrelated to the patient's actual condition, such as an over-recommendation of anti-infective medications. Other models, lacking effective filtering of longitudinal visit records, perform less effectively in capturing dynamically adjusted medications. While DAINet achieves a reasonable Correct value, its higher Unseen value suggests a tendency to recommend unnecessary auxiliary medications, increasing potential risks. RAREMed achieves relatively low Unseen values, indicating its strength in avoiding incorrect predictions. However, its lower Correct values reflect a limited ability to capture complex drug combinations, making it unsuitable for scenarios requiring dynamic adjustments or acute-phase treatments. CausalMed exhibits a performance trend similar to RAREMed across different visit numbers. Although CausalMed shows a slight advantage over RAREMed in terms of correct predictions, it still falls short of RaVSNet in both accuracy and safety. This indicates that, while CausalMed benefits from causal modeling strategies, it does not fully leverage cross-visit information as effectively as RaVSNet, resulting in inferior overall performance.

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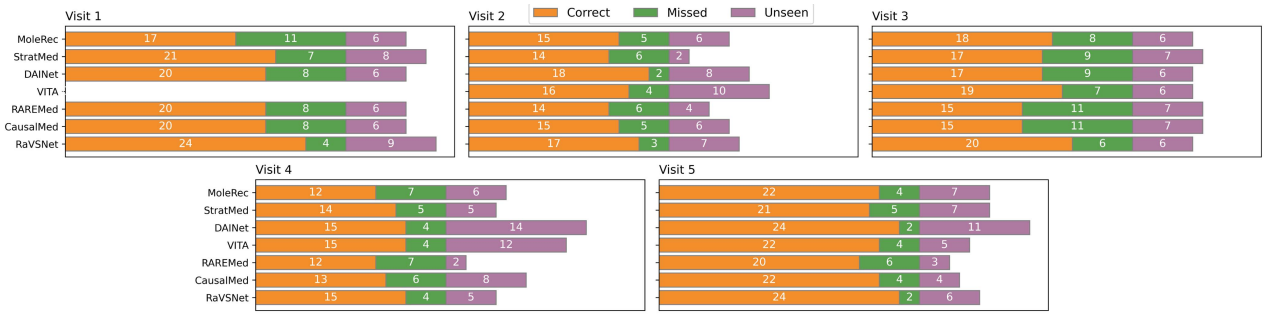


Fig. 7. Drug recommendation comparison across visits and models.

VI. Conclusion

This paper presents RaVSNet, a relevance-aware visit similarity network for drug recommendation, along with a general pretraining framework. By jointly modeling longitudinal and transversal visit similarities and leveraging medical relevance knowledge, RaVSNet improves both the accuracy and safety of medication recommendations. Extensive experiments on MIMIC-III and MIMIC-IV demonstrate that RaVSNet achieves state-of-the-art performance across multiple evaluation metrics. However, RaVSNet exhibits relatively high computational complexity due to its integration of both longitudinal and transversal visit similarity modules, as well as a relevance-aware alignment mechanism. The computational complexity is similar to COGNet [31] and VITA [20]. This design is motivated by the need to enhance accuracy and medication safety, particularly in sparse or high-risk patient scenarios. To explore ways of improving efficiency, we conducted a preliminary experiment using a distilled student version of RaVSNet, which retains only the visit representation and bias correction modules. The student model adopts a much simpler structure that is comparable in complexity to models like SafeDrug [41], while achieving performance close to the original RaVSNet. Notably, it still outperforms all baselines in most evaluation metrics. These results suggest that RaVSNet is amenable to future compression and deployment-oriented optimization without compromising its core advantages.

For future work, we aim to enhance interpretability by incorporating knowledge graphs and extracting reasoning paths between clinical concepts and medications. We also plan to reduce computational overhead through approximate similarity retrieval and knowledge distillation. To improve generalizability, we will evaluate RaVSNet on more diverse datasets (e.g., eICU, All of Us) and integrate multimodal patient information, such as genomic data and social determinants of health. Finally, to support adaptation across heterogeneous healthcare systems, we intend to incorporate missing value imputation and metadata-aware confidence modeling, leveraging RaVSNet's modular and data-agnostic architecture.

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